

Chapter 1 The Pythagorean Theorem

1.1 Exploring the Pythagorean Theorem

In a right triangle, the sum of the squares of the two legs equals the square of the hypotenuse. If the legs have lengths a and b and the hypotenuse has length c , then $a^2 + b^2 = c^2$. This is the famous Pythagorean theorem.

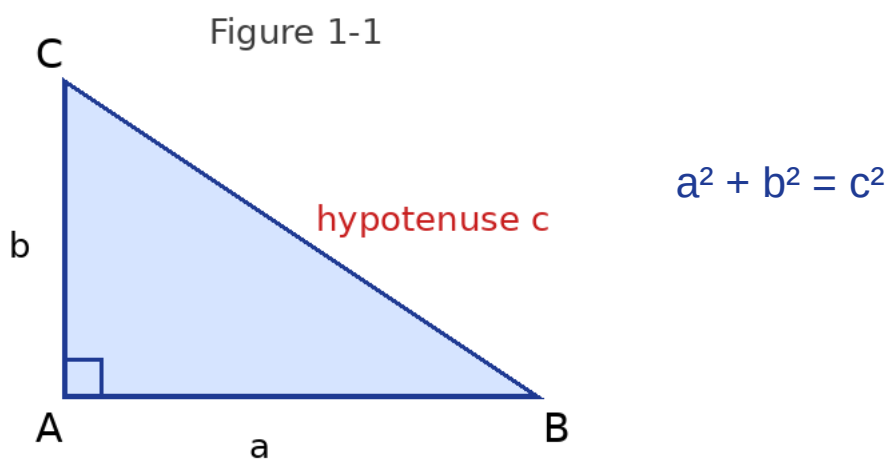


Figure 1-1 Side lengths in a right triangle

Example 1 The legs of a right triangle are 3 cm and 4 cm. Find the length of the hypotenuse.

Solution: By the Pythagorean theorem, $c^2 = 3^2 + 4^2 = 9 + 16 = 25$, so $c = 5$ cm.

Theorem: In a right triangle the sum of the squares of the legs equals the square of the hypotenuse.

Exercises 1.1

1. In right triangle ABC, $\angle C = 90^\circ$, $AC = 6$ and $BC = 8$. Find AB.

2. The hypotenuse of a right triangle is 13 and one leg is 5. Find the other leg.

3. Decide: is a triangle with sides 7, 24 and 25 a right triangle?

Think: as shown in the figure, four congruent right triangles form a large square with a small square in the middle. Use areas to prove the Pythagorean theorem.

